

# *Voice Activity Detection Using Novel Teager Energy Based Band Spectral Entropy*

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**Abstract**—there are many features proposed in the literature for voice activity detection (VAD). Shen et al. [20] first used a spectral entropy-based feature to detect regions of speech spurts under noisy conditions. However, VAD employing this feature was unreliable when the noise level greatly exceeds the speech level. To improve the performance of spectral entropy based VAD under low signal-to-noise ratios (SNRs), spectrum of a signal over a frame is divided into sub bands and spectral entropy is computed over these bands. Later, these spectral entropies are weighted and summed to obtain the entropy. Based on the amount of noise in each band, weights were found empirically. This approach was named as banded spectral entropy (BSE) [21]. In [24], deviation threshold computed from approximate ramp line and the sorted spectral coefficients of the band are adopted to decide useful/useless bands. In this paper, we propose a novel Teager Energy Band Spectral Entropy (TE\_BSE) feature for VAD. Here, we carryout enhancement of spectral peaks employing Teager energy of each frequency transformed speech frame. This is followed with dividing of spectrum into sub bands and entropy computation over each band. The summing of entropy from each useful band is done to get TE\_BSE feature. We identify useful/useless bands following [24]. Later, we present the performance of our proposed VAD in terms of probability of detection ( $P_D$ ), probability of false alarm ( $P_{FA}$ ) and probability of error under different noises and SNRs. Finally, from the VAD results on real-world sample, proposed VAD outperforms statistical based VAD by Sohn et. al. [8] with improved  $P_D$  not at the cost of increase in  $P_{FA}$ .

**Keywords**—Voice Activity Detection; Spectral Entropy; Band Spectral Entropy; Teager Energy Operator;

## I. INTRODUCTION

A process of marking the boundaries or classifying between speech and non-speech (silence, background noise etc.) segments of a digital speech signal is popularly known as voice activity detection (VAD). Speech/non-speech detection as a preprocessing step, enhances the performance of applications including robust speech recognition [1,2], discontinuous transmission [3,4], real-time speech transmission on the Internet [5] or combined noise reduction and echo cancellation schemes in telephony [6,7]. Speech/non-speech classification task is not as trivial as it appears; indeed, most VAD algorithms fail when the background noise increases. Numerous researchers have proposed different schemes for detecting speech regions in a noisy speech signal [8-11].

The different VAD methods include those based on energy thresholds [13], pitch detection [16], spectrum analysis [15], zero-crossing rate [3], periodicity measure [17], higher order statistics in the Linear Predictive Coding (LPC) residual domain [18] or combinations of different features [3,4,19]. Shen et al. [20] first used a spectral entropy-based parameter to detect regions of speech spurts under noisy conditions. However, their method is unreliable when the noise level greatly exceeds the speech level. Additionally, the spectral entropy relies on the variance of spectral magnitude to distinguish speech from noise, but the variance of spectral magnitude in turn strongly depends on the nature of noise.

One can improve the robustness of spectral entropy by exploiting banded structure of speech spectrogram and discarding bands dominated by noise, thereby enhancing the banded nature of speech spectrogram. Spectral entropy thus computed is called the banded spectral entropy (BSE) [21]. However, BSE does not sufficiently specify the organization of the banded nature on the spectrogram. Therefore, in [21], Wu and Wang have proposed a set of weights to compensate for this drawback. These weights have been found empirically by the authors [21] to sufficiently characterize the speech signals rather than by simply computing spectral entropy of each sub-band [20]. Although the weighted BSE remains a good feature parameter, the detection sometimes fails at very low SNRs. The question that needs to be addressed is the following: *How is one to discard the noisy bands and hence, effectively retain only the useful bands?*

In [22], it was shown that the number of noisy bands (or useful bands) is related to the background noise level. A Minimum Mel-scale frequency Band (MiMSB) parameter proposed in [23] has been used in [21] to estimate the varying noise levels by adaptively choosing one band with minimum energy. A Normalized Minimum Band Energy (NMinBE) parameter has been proposed to accurately decide the number of useful bands. The authors in [21] have used the number of useful bands empirically derived from the knowledge of NMinBE. Further, it has been assumed in [21] that a short window of non-speech interval exists prior to the commencement of the input signal and the mean and variance of the logarithmic Adaptive BSE (ABSE) has been initialized on the estimates computed from this window. Successively, the relative mean and variance of the logarithmic ABSE values are updated during non-speech periods.

In contrast, our proposed band selection scheme is easy to setup and has too less empirical parameters as part of the initialization. This scheme exploits the banded structure available in noisy speech spectrum and selects those sub-bands where speech dominates over noise. We examine the banded structure on speech spectrogram to detect speech/non-speech regions as used in [21], but differ from it in terms of selection of useful/useless bands. Our band selection scheme depends not exclusively on the statistical behaviour of speech, but also on speech contaminated by noise in sub-bands. Specifically, we exploit the well-known fact that the formants play a major role in modelling speech signals in a compact way, and that the first three formants are sufficient to synthesize intelligible speech. Therefore, the detection of speech/non-speech is equivalent to identifying the presence or absence of formants.

In this paper, we adopt band spectral entropy (BSE) in [21] to compute the entropy. However, our approach is different on two counts (i) Selection of useful/useless bands in a spectrum employing deviation between coefficients that approximate ramp line (ARL) and sorted spectral coefficients (ii) Enhancement of spectral peaks employing Teager energy operator (TEO) on the signal spectrum such that the spectral peaks are enhanced. This is followed with the smoothening of present spectral estimates with past spectral estimates. Further, our work deviates from [24] on second count. In section, II we discuss spectral peak enhancement and smoothening employing TEO and later computation of BSE. We name this new feature as Teager energy band spectral entropy (TE\_BSE). Section, III presents theory behind the selection of useful/useless bands proposed in [24]. In section, IV we present experimental setup in extracting TE\_BSE feature followed with performance analysis this feature under (i) Stationary noise (ii) Non-stationary noise and (iii) Non-stationary heavy noise environments. Finally, we present the VAD output employing proposed feature and compare with statistical based VAD by Sohn et. al. [8]. This followed with concluding remark and some vistas for future.

## II. BAND SPECTRAL ENTROPY

We now review the procedures in [20, 21] in some more detail. The entropy measures the randomness of the signal  $x(\cdot)$  with the defined probability density function  $f(x)$  as

$$H(x) = - \int f(x) \log(f(x)) dx. \quad (1)$$

Shen et al. [20] employed it to discriminate speech from non-speech signals as part of ED. The speech signal is observed over a frame of length,  $N$ . The Short-Time Fourier Transform (STFT) of  $l$ th frame,  $s(n, l)$ , is given by

$$S(k, l) = \sum_{n=0}^{N-1} w(n) s(n, l) \exp\left(-\frac{j2\pi kn}{N}\right), \quad k = 0, \dots, N-1, \quad (2)$$

where  $|S(k, l)|$  represents the spectral magnitude of the  $k$ th frequency bin of the  $l$ th frame,  $N$  being the total number of frequency bins and  $w(n)$ , a Hamming window. Excluding the symmetry and squaring each of  $|S(k, l)|$ , let

$$S_{msq}(k, l) = |S(k, l)|^2, k = 0, \dots, \frac{N}{2} - 1. \quad (3)$$

Then, the probability associated with each  $S_{msq}(k, l)$  component  $P(j, l)$  can be estimated by normalizing

$$P(j, l) = \frac{S_{msq}(k, l)}{\sum_{k=0}^{\frac{N}{2}-1} S_{msq}(k, l)}, 0 \leq j \leq \frac{N}{2} - 1. \quad (4)$$

The corresponding spectral entropy over a frame is

$$H(l) = \sum_{j=0}^{\frac{N}{2}-1} P(j, l) \log(1/P(j, l)). \quad (5)$$

Spectral entropy depends on the variation of the spectral energy but not on the amount of spectral energy. Consequently, spectral entropy, and hence the performance of VAD, are robust to the changing noise levels [20]. However, the magnitude associated with each point in the spectrum is affected by noise and, therefore, VAD performance would degrade severely at low SNRs. Further, the observation in [21] about 'effect of noise in the spectrum' was primarily restricted to a few sub-bands instead of a full band. This naturally suggests multiband analysis and necessitates the entropy computation for each of the sub-bands [21]. The sub-band energy of a given frame is computed as follows:

$$E_{SBE}(m, l) = \sum_{k=(m-1)*(N/2N_b)}^{(\frac{N}{2N_b})-1+(m-1)*(N/2N_b)} S_{msq}(k, l), \quad 1 \leq m \leq N_b, \quad (6)$$

where  $N_b$  is the total number of sub-bands for each frame ( $N_b = 32$ ) and represents the energy of the sub-band. Again, we define the probability associated with each sub-band as

$$P_{SBE}(m, l) = \frac{E_{SBE}(m, l)}{\sum_{k=1}^{N_b} E_{SBE}(k, l)}, 1 \leq m \leq N_b. \quad (7)$$

The BSE parameter,  $H_{SBE}(l)$ , is now computed as

$$H_{SBE}(l) = \sum_{m=1}^{N_b} P_{SBE}(m, l) \log(1/P_{SBE}(m, l)). \quad (8)$$

Further, in [21], a set of weights is proposed to compensate for this drawback. These weighting factors are found empirically to characterize speech signals more effectively than by simply computing spectral entropy of each sub-band. Although the weighted BSE remains a good feature, the detection sometimes fails at very low SNRs. Therefore, it is required to discard the noisy bands and retain others.

### A. BSE using Teager Energy Operator

Formant tracking using the Teager energy operator (TEO) has been proposed by many researchers [25-28]. In [28], multiband approach along with TEO is employed in formant tracking. We are motivated by the efficacy of TEO in formant estimation and have adopted it in our approach. A discrete version of TEO on a time domain signal,  $x(n)$  is given by

$$E_{Teager}(x(n)) = x^2(n) - x(n-1)x(n+1), \forall n. \quad (9)$$

To enhance peaks in the spectrum, and thereby to track the formants, we employ TEO on  $S_{msq}(k, l)$ ,  $k = 0, \dots, -1$ , in (3). This results in  $S_{Tmsq}(k, l)$ . However, this enhancement comes at the cost of enhancing spurious peaks in  $S_{msq}(k, l)$  due to the difference operation in TEO (refer (9)). To derive the benefits of TEO and minimize spurious peaks, we use convex combination of  $S_{msq}$  and  $S_{Tmsq}$  as follows.

$$E(k, l) = \alpha S_{msq}(k, l) + (1 - \alpha) S_{Tmsq}(k, l), \quad k = 0, \dots, (N/2) - 1. \quad (10)$$

Here,  $0 \leq \alpha \leq 1$ . To find the probability associated with  $l$ th frame, we use (4) and replace  $S_{msq}(k, l)$  by  $E(k, l)$  as

$$P_T(j, l) = \frac{E(k, l)}{\sum_{k=0}^{N/2-1} E(k, l)}, \quad 0 \leq j \leq \frac{N}{2} - 1. \quad (11)$$

The corresponding spectral entropy is computed as follows:

$$H_T(l) = \sum_{j=0}^{N/2-1} P_T(j, l) \log\left(\frac{1}{P_T(j, l)}\right), \quad 0 \leq j \leq \frac{N}{2} - 1. \quad (12)$$

We observe that the dominance of formants in sub-bands introduces sparseness (presence of structure) and the absence of formants or noise contamination introduces denseness (absence of structure). We attempt to detect formants by utilizing banded structure of speech signals [21]. From  $E(k, l)$ , we compute the sub-band energy, sub-band probability and associated entropy using (6), (7) and (8). Thus,

$$E_{SBE}^T(m, l) = \sum_{k=(m-1)*(N/2N_b)}^{(N/2N_b)-1+(m-1)*(N/2N_b)} E(k, l), \quad 1 \leq m \leq N_b, \quad (13)$$

$$P_{SBE}^T(m, l) = \frac{E_{SBE}^T(m, l)}{\sum_{k=1}^{N_b} E_{SBE}^T(k, l)}, \quad 1 \leq m \leq N_b \quad (14)$$

and

$$H_{SBE}^T(l) = \sum_{m=1}^{N_b} P_{SBE}^T(m, l) \log(1/P_{SBE}^T(m, l)). \quad (15)$$

### III. SELECTION OF USEFUL/USELESS SPECTRAL BANDS

In [20], weighted BSE is supported with useful/useless band selection using NMinBE parameter under low SNRs. We differ from [20, 21] in the selection of useful/useless sub-bands. Here, we examine each frame by equally partitioning  $P_T(j, l)$ ,  $0 \leq j \leq \left(\frac{N}{2}\right) - 1$ , into  $N_b$  sub-bands as  $P_T^b(m, l)$ , where  $b = 1, \dots, N_b$ , and  $m = \left(\frac{(b-1)N}{2N_b}\right), \dots, \left(\frac{bN}{2N_b}\right) - 1$ . The presence/absence of structure in sub-bands is equivalent to the presence/absence of formants in the sub-bands. We argue that the presence of formant amounts to sparseness and that their absence, due to the presence of noise, yields denseness. This provides a basis to measure sparseness or denseness via the deviation of an approximate ramp line (ARL) [24] from the sorted  $P_T^b(m, l)$  coefficients. The sorted  $P_T^b(m, l)$  coefficients are henceforth denoted as  $\tilde{P}_T^b(m, l)$ . We connect the first

coefficient of  $\tilde{P}_T^b(m, l)$  to the last coefficient to generate ARL. Subsequently, we sample ARL with uniform intervals between samples. Here, ARL is computed for each sub-band in the frame of a given speech signal.

We observe that for the sub-band, the  $\tilde{P}_T^b(m, l)$  coefficients are closer to ARL when non-speech/noise dominates over speech and that they are relatively removed from it otherwise. Closeness to ARL is computed by taking sum of square difference between samples of ARL and  $\tilde{P}_T^b(m, l)$  coefficients as

$$CL^b(l) = \sum_{m=\frac{(b-1)N}{2N_b}}^{\frac{bN}{2N_b}} (ARL^b(m, l) - \tilde{P}_T^b(m, l))^2. \quad (16)$$

We present some features of the proposed scheme.

- When speech formants dominate over noise in a sub-band, the deviation of  $\tilde{P}_T^b(m, l)$  coefficients from ARL is higher than otherwise. This accounts for correct performance.
- When noise dominates over speech formants in all the sub-bands, speech sub-bands may be identified as non-speech, thus resulting in Missed Detection (MD). This occurs when speech is completely masked by the noise under very poor SNR conditions.
- Noise induced spectral peak in the sub-bands results in false alarm (FA) in the event of sub-bands having no speech signal. This occurs when a non-stationary narrow band noise in non-speech region presents itself a peak within sub-band. Then, any appreciable deviation generated by this noisy spectral peak away from ARL will result in a non-speech packet being detected as a speech packet.

The nature of ARL and  $\tilde{P}_T^b(m, l)$  coefficients under additive babble noise during speech/non-speech frames for the SNRs, 0, 5 and 15dB is presented through Fig. 1 – Fig. 3. Here, the sub-band  $b$  is fixed and  $\tilde{P}_T^b(m, l)$  coefficients from (i) Speech plus babble noise frame; and (ii) Babble noise only frame are used to obtain ARL and plot of sorted coefficients of  $\tilde{P}_T^b(m, l)$ , respectively. Conclusions drawn from these plots are summarized as follows:

- When SNR is at 0 dB, the difference in the range of  $\tilde{P}_T^b(m, l)$  for speech and non-speech frames is not so pronounced. However, large deviation of  $\tilde{P}_T^b(m, l)$  from ARL can be noticed for the speech frame than for non-speech frame. This helps in deciding *speech/non-speech* sub-bands in a frame and to the computation of SBE when a sub-band is declared as *speech*. As SNR increases,  $\tilde{P}_T^b(m, l)$  range increases/decreases for speech/non-speech frames as also the deviation from ARL.

A Deviation Threshold (DT) is selected empirically to achieve minimum MD and restrict FA to an allowable range. Each sub-band is checked for the presence or absence of formants based on the deviation achieved and compared with the DT. This refined adaptive band selection scheme will adapt quickly to change in signal and noise conditions. Thus,

the proposed approach has less parameters to setup arbitrarily than in [21]. Threshold for BSE is initialized assuming first 200ms as non-speech region. We differentiate speech regions from non-speech regions using the BSE feature computed from useful sub-bands. A simple hangover scheme is employed to minimize fast switching between speech and non-speech and vice versa.

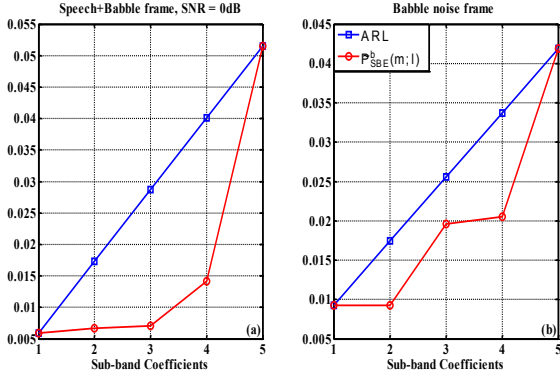


Fig. 1. Plot showing ARL and  $\tilde{P}_T^b(m, l)$  under babble noise with 0dB SNR (a) Speech plus noise (b) noise only frame.

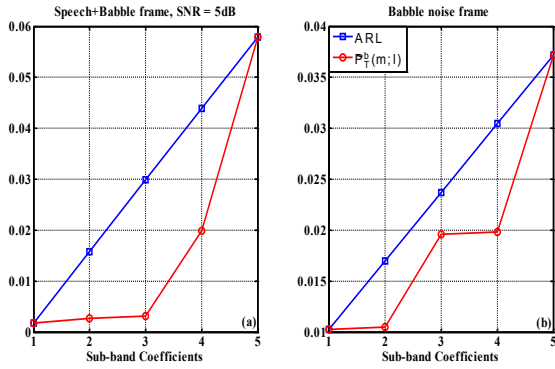


Fig. 2. Plot showing ARL and  $\tilde{P}_T^b(m, l)$  under babble noise with 5dB SNR (a) Speech plus noise (b) noise only frame.

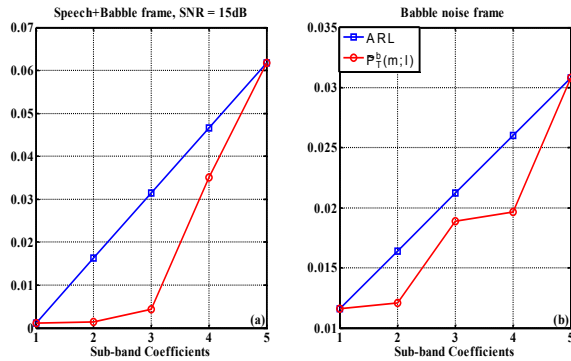


Fig. 3. Plot showing ARL and  $\tilde{P}_T^b(m, l)$  under babble noise with 15dB SNR (a) Speech plus noise (b) noise only frame.

#### IV. EXPERIMENTAL SETUP AND RESULT ANALYSIS

We evaluated our VAD scheme by considering clean and noisy speech signals that are sampled and digitized. Sampling frequencies of 8 and 16 kHz are considered in our evaluation.

Then, sampled and digitized speech signals are framed such that the successive frames do not overlap. Here the frame length is chosen to accommodate 320 speech samples. The number of sub-bands,  $N_b$ , is fixed to 32. In computing  $E(k, l)$  using (10), we set  $\alpha$  to be 0.3. To simulate the noisy speech with SNRs from 0 – 40 dB, speech signal is added with 13 different noises: Babble, F16, Tank, Factory, Car, Subway, Fan, Pink, HFchannel, Machine Gun, Destroyer Engine, Military Vehicle, Jet Cockpit and White.

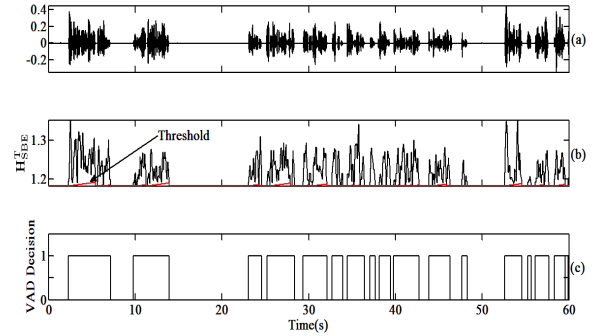


Fig. 4. Result of VAD under clean signal (a) Speech signal (b) Band Spectral Entropy and Threshold (c) VAD Decisions (Speech = 1; Non-speech = 0).

The SNR is varied in steps of 5dB. The results of VAD using proposed scheme under clean and noisy conditions are presented (refer Fig. 4). Here, Fig. 4(a) is speech sample under clean conditions. Figure 4(b) shows  $H_{SBE}^T$  and the threshold computed for the speech signal in Fig. 4(a). The corresponding

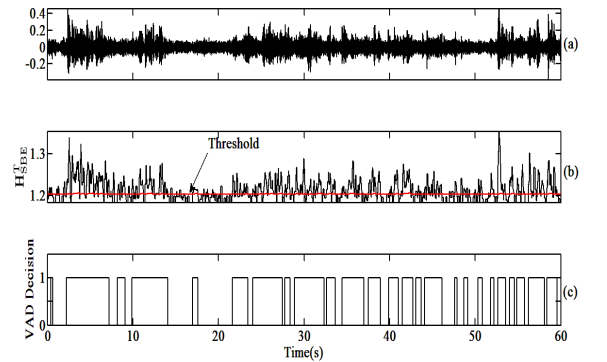


Fig. 5. Result of VAD under Babble noise with SNR = 0dB (a) Speech signal (b) Band Spectral Entropy and Threshold (c) VAD Decisions (Speech = 1; Non-speech = 0).

VAD decisions are in Fig. 4(c). In Fig. 5 – Fig. 7, we added Babble, Jet Cockpit and Subway noises to the speech in Fig. 6(a) to simulate respective noisy speech signals where the SNR is maintained at 0 dB. Parts (c) of Fig. 5 – Fig. 7 show VAD decisions with the proposed approach. We observe from Fig. 5(c) that in the presence of speech-like noise, such as Babble, FA errors are more pronounced than MD errors under low SNR conditions. However, the VAD decisions presented in Fig. 6(c) and Fig. 7(c) are for non-speech like noises such as Jet Cockpit and Subway noises where it can be observed

that FA and MD errors are minimal for the same SNR setup. This is attributed to the robustness of the proposed speech/non-speech band selection scheme under additive non-speech like noises as against Babble noise.

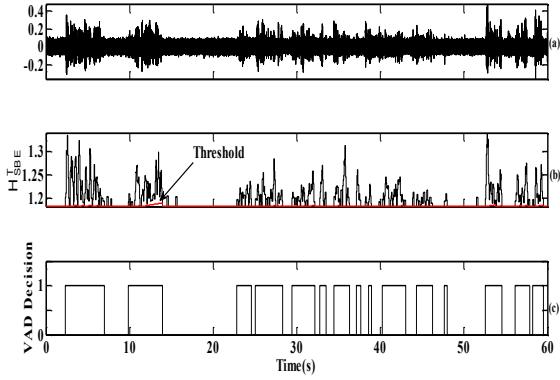


Fig. 6. Result of VAD under Jet Cockpit noise with SNR = 0dB (a) Speech signal (b) Band spectral entropy and threshold (c) VAD Decisions (Speech = 1; Non-speech = 0).

We evaluate the proposed scheme considering more than 46 speech samples from the *Switchboard Corpus*, a corpus of spontaneous conversations by multiple speakers over telephone [29]. In the performance evaluation that follows, we compare VAD results from our scheme with the manually segmented speech/non-speech information provided with the database. The results are presented for the SNRs from -10 to 40dB. Here, speech with additive noises is broadly classified under three categories: (i) Non-stationary Noisy Speech (NsNS) (ii) Non-stationary Heavy Noisy Speech (NsHNS); and (iii) Stationary noisy speech (SNS).

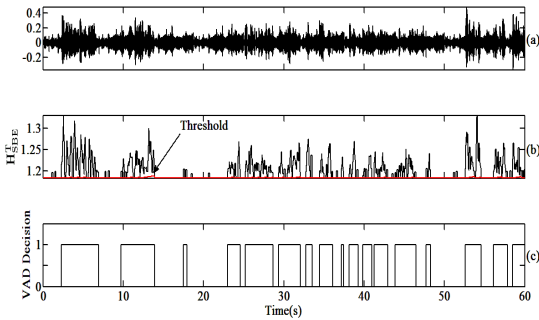


Fig. 7. Result of VAD under Subway noise with SNR = 0dB (a) Speech signal (b) Band spectral entropy and threshold (c) VAD Decisions (Speech = 1; Non-speech = 0).

Analysis of the performance of our scheme under these three noisy conditions averaged over the number of speech samples is presented as follows:

- $P_D^S$  represents the ratio of the number of speech packets correctly recognised to the total number of speech packets. In Fig. 8(a)  $P_D^S$  is more than 0.8 for the SNRs greater than or equal to 0 dB. Under NsHNS conditions (see Fig. 8(b)), we achieve similar performance as in the Fig. 8(a) except

for the case of Destroyer Engine noisy conditions. However for SNS conditions as presented in Fig. 8(c), similar performance of Fig. 8(a) is achieved except for the case of additive Car and Fan noisy conditions. This is due to the effect of these noises on the banded structure of speech; consequently, DT fails to detect the speech bursts.

- However, the corresponding Probability of False-alarm ( $P_{FA}$ ) presented in Fig. 9(c) is lower than the achieved  $P_{FA}$  for NsNS and NsHNS conditions (see Fig. 9(a) and Fig. 9(c)). The trade-off is significant.
- Probability of Error, ( $P_E$ ), combines both the errors  $P_{FA}$  and  $P_{MD}$ --upon multiplying with the prior probabilities, i.e. probability of speech packets and probability of non-speech packets for a given test speech sample. Thus,  $P_E$  represents the total error vis-à-vis speech - non-speech detection and the goal is to achieve lower  $P_E$  under different noisy conditions. Fig. 10 presents  $P_E$  for the three noisy conditions. Under SNS conditions,  $P_E$  is lower than 0.1 for SNRs greater than 0 dB, except the Car and Fan noises. This is due to higher  $P_{MD}$  even though  $P_{FA}$  is less than 0.2. For NsNS & NsHNS conditions,  $P_E$  is less than 0.2 for SNRs greater than 0 dB except Babble, Destroyer Engine and Military Vehicle noises, as shown in Fig. 10(a) and Fig. 10(b). We know that the Babble noise is speech like and as expected,  $P_{FA}$  is higher, and hence, higher  $P_E$ .

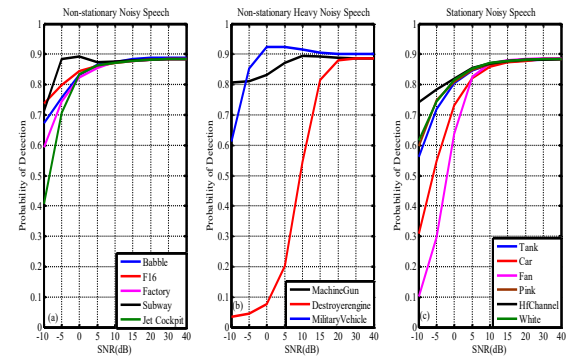


Fig. 8. Probability of speech detection ( $P_D^S$ ) vs. SNRs in dB (a) Non-stationary noisy speech (b) Non-stationary heavy noisy speech (c) Stationary noisy speech.

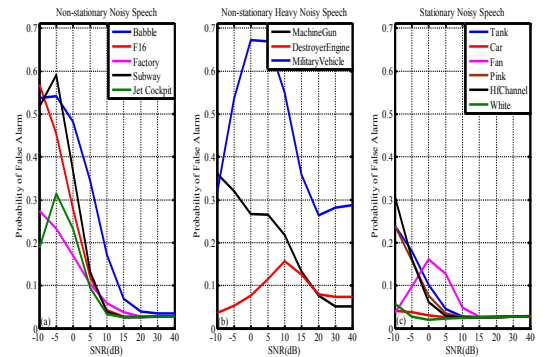


Fig. 9. Probability of False-alarm ( $P_{FA}$ ) vs. SNRs in dB (a) Non-stationary noisy speech (b) Non-stationary heavy noisy speech (c) Stationary noisy speech.



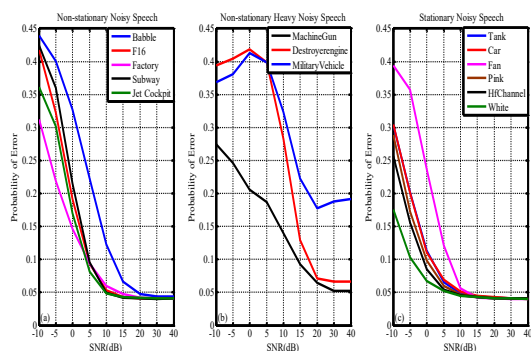


Fig. 10. Probability of Error ( $P_E$ ) vs. SNRs in dB (a) Non-stationary noisy speech (b) Non-stationary heavy noisy speech (c) Stationary noisy speech.

We present VAD of our scheme in Fig. 11 for a real world speech sample having background noise and low fidelity. Result thus obtained is compared with the popular scheme proposed by Sohn et. al. [8]. For the  $P_D^S$  under both schemes are similar with minimal  $P_{MD}$  in favour of our scheme. It is important to note that this performance also comes with minimal  $P_{FA}$ . However, it is not true for VAD in [8]. Further, perceptual and statistical tests on VAD results corroborate good performance of our scheme over [8].

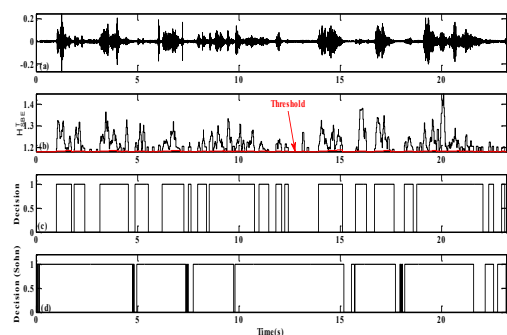


Fig. 11. VAD for a real world speech sample (a) Non-stationary noisy speech (b) Band spectral entropy and threshold (c) VAD Decision using proposed scheme (d) VAD decision using scheme proposed by Sohn et. al. [8].

## CONCLUSION

In this paper, we proposed a novel Voice Activity Detection (VAD) using Teager Energy based Band Spectral Entropy (TE\_BSE). We segmented the input speech signal into non-overlapping frames. For each frame, we compute the entropy via normalized frame signal energy obtained for each sub-band. However, this parameter was seen to be sensitive to noise. We recognized that the signature of speech signal is embedded in a formant structure. We employed the Teager Energy Operator (TEO) to enhance formant peaks of the sub-band spectrum. This comes at the cost of enhancing high frequency regions of the spectrum under low SNRs. To exploit the benefits of normalized signal energy and TEO, we employed them under convex combination. Using the approach in [24], to obtain deviation threshold to declare each sub-band as useful or useless. Useful sub-bands are used to compute

spectral band entropy. We differentiated *speech regions* from *non-speech regions* using the SBE feature computed from useful sub-bands. Finally, we evaluated the proposed scheme under simulated noisy (NsNS, NsHNS and SNS) and real world conditions to assess the robustness in providing accurate VAD decisions. The enhanced performance of the proposed schemes under varying noise conditions as claimed through the simulation studies is actually borne out in the implementation on real world speech signals.

The schemes proposed and analyzed in this work may readily be gainfully employed for cleaning up several speech signals when contaminated by various types of noise from the environment. One of the very promising applications of the VAD studied here can be in speech compression for transmission in the context of audio conferencing over the VoIP such as in Skype.

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